Q-TRACE · November 24, 2026 · Final experimental findings

One-page summary · meeting 6 of 7 · one-time sealed test of October 8, 2026

Chronology. The sealed test was opened once on Oct 8, 2026 (21:28 UTC). The manuscript and capstone report were drafted the same day. Before presenting, update slide 5 with any author decisions made since.

Purpose. To report the held-out evaluation, compare the rule with learned comparators, show the reliability–resource trade-off, state the statistical limits, and outline the publication path.

Key numbers (16 unseen models, 4 unseen structures).

Item	Result	Source
Rule R_thr, continuous noise	material errors 2.0% [1.4, 2.6]; Brier regret −0.03%; +0.6% shots vs Oracle; −3.4% vs always-combined	`final_policies_5c_test.csv`
Rule, categorical noise	8.8% (registered bound 10%)	`final_policies_5b_test.csv`
Best learned (gradient boosting)	2.1% material; difference +0.0005 [−0.016, +0.027]; any disagreement 6.5% vs 21.9%	`final_paired_vs_rthr_5c_test.csv`
Always-combined	5.9% material = 1 gate-free setting in 17	`final_policies_5c_test.csv`, `p5c_design.csv`
Hypotheses	H1–H4 supported	`final_metadata.json`

Limits to state.

- Only 4 held-out structures, so the intervals are likely too narrow.
- H2 is a non-superiority result on material errors, not proof that learning cannot help.
- On raw agreement with the Oracle, learned models were closer.
- The evidence is simulation only, at 4 qubits, with a gate probe nearly blind to thermal noise.

Contribution.

1. Noise damaged calibration, not discrimination; mitigation repaired it, and extra shots did not.
2. A leakage-safe benchmark of 64 models.
3. A pre-registered test in which a simple probe rule performed competitively with learned comparators.

Publication path (authors decide). Nothing has been submitted. Recommended route: arXiv, then TMLR. Alternatives: IEEE QCE27 or Quantum Machine Intelligence (`deliverables/submission/VENUES.md`).

Slide	Content	Speaker (proposed)
1	Title and message	Tushar
2	The one-time sealed evaluation	Tushar
3	Rule vs learned comparators	Tushar
4	Trade-offs and limits	Mayank
5	Contribution and publication path	Mayank

Needs confirmation.

- The presenter split.
- The meeting date and format.
- Any venue or author decisions to add to slide 5.